



Optimum Material Selection for Cryogenic Tanks: An Integrated Criteria Importance Through Intercriteria Correlation–Combinative Distance-based Assessment Approach

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Abstract: Choosing optimal materials for cryogenic storage systems is a challenging intelligent decision-making task with several competing requirements. To determine the best material for producing cryogenic tanks used in the transportation of liquid nitrogen, this study proposed an integrated multi-criteria decision-making (MCDM) framework that combined the Criteria Importance through Intercriteria Correlation (CRITIC) method with the Combinative Distance-based Assessment (CODAS) method. Seven technical performance criteria, including toughness index, yield strength, density, Young’s modulus, thermal expansion, thermal conductivity, and specific heat were adopted to assess seven potential materials. By considering both contrast intensity and intercriteria correlation, the CRITIC technique could scientifically establish criteria weights while reducing subjective bias. The options were ranked using the CODAS approach according to their Euclidean and Taxicab distances from the negative ideal solution. The findings demonstrated that density had the greatest management weight when it came to sustainable design. Therefore, aluminium 5052-O was the most appropriate material for cryogenic tank applications out of all the alternatives under investigation. The proposed CRITIC–CODAS framework, a dependable intelligent decision-support tool for strategic material selection in advanced manufacturing and engineering management contexts, exhibits robustness, transparency, and computing efficiency.

Keywords: Multi-criteria decision-making; Material selection; Criteria Importance through Intercriteria Correlation; Combinative Distance-based Assessment; Cryogenic tanks

1 Introduction

Over the past few years, several conventional materials have been replaced by modern materials due to increase in the demand for materials and their low weights. The main reason behind the selection of materials in engineering design is that, material selection does not just need information about different electric, magnetic, physical, mechanical, manufacturing and chemical properties but it also requires the cost of material, shape of the product, effect of the environment, characteristics of performance, materials availability, and consideration of the design. The relationship between dissimilar selecting criteria affects the whole process of selection. There are many different types of methods to tackle the problems arising from the decision making in material selection; these approaches involve the screening of materials and selection methods [1]. The main component of Multi-criteria decision-making (MCDM) theory is the use of computational techniques to evaluate and choose the best option among the alternatives and aimed at achieving the intended result by incorporating a nu criteria and an order of preference. It can derive

the best solution to problems encountered in different fields. Users' experiences cannot determine among the many different parameters under consideration. The application provides a ranking result that is based on the selected criteria, their corresponding values, and assigned weights. MCDM technique is a practically useful tool for making decisions with respect to material selection. Different MCDM methods have been devised by researchers to solve the crucial problems involved in this process; MCDM methods which have been in place for analysis include the Analytic Hierarchy Process (AHP) and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), Data Envelopment Analysis (DEA) and Preference Ranking Organization Method for Enrichment of Evaluations (PROMETHEE), etc. The application areas of MCDM method encompass automotive, agriculture, energy and power, chemical, maritime, health, manufacturing, and construction [2].

Numerous materials including metallic and non-metallic engineering alloys exist in the contemporary world. The materials that have been used so far in cryogenic tanks are Al 2024 -T6 which has great machinability and good finishing of the surface capability and, also has very highly strengthened material, especially for workability. On the other hand, the adoption of Aluminium 5052-0 is based on its better workability, quite good corrosion resistance, and good weldability. Another material that finds wide application for cryogenic tanks is stainless steel 301 that shows good strength, great ductility, and is one of the best corrosion resistant. Material optimization for engineering applications has been investigated by several researchers. For instance, Sreejesh and Manoj [3] employed finite element analysis (FEA) to compare silumins with aluminium alloy pistons. According to their research, silumin pistons reduced weight by about 17% and volume by 10% when compared with aluminium alloys, thus suggesting that this material might be produced in large quantities at a reasonable cost. Anojkumar et al. [4] applied MCDM techniques to address material selection for piping applications in the sugar industry. Four decision-making methods were used to evaluate five stainless steel grades (J4, JSLAUS, J204CU, 409M, and 304) based on multiple criteria, including tensile strength, hardness, cost, corrosion resistance, wear, and yield strength. The results indicated that grade 304 Stainless Steel exhibited superior corrosion resistance compared with the other alternatives. Chatterjee et al. [1] applied multiple MCDM techniques, including Complex Proportional Assessment (COPRAS), EVALuation of MIXed Data (EVAMIX), TOPSIS, and VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), to evaluate materials for cryogenic tank applications, and identified SS301-FH as the optimal choice. Rao and Davim [5] developed a decision-making framework integrating AHP and TOPSIS for material selection in cryogenic storage tanks used for liquid nitrogen transport. Their findings identified SS301-FH as the optimal material. Dehghan-Manshadi et al. [6] presented the weighted properties method (WPM) for material selection and evaluated several candidate materials for cryogenic applications. Their results indicated that stainless steel 301 exhibited the best overall performance, followed by titanium alloy, Inconel 718, stainless steel 310, aluminium alloy, and brass. Roth et al. [7] applied multi-attribute utility analysis using subjective probability assessment for material selection in engineering applications. Their approach was demonstrated for automobile camshaft and aircraft gas turbine exhaust duct materials, thus identifying microalloyed steel and powder-forged steel as the most suitable options. Yelamasetti et al. [8] investigated the weldability and mechanical behaviour of dissimilar joints between AL5052 and AL7075 produced using the Gas Tungsten Arc Welding (GTAW) process. The study employed ER4043 and ER5356 filler wires and the results indicated that successful welding of the dissimilar combination was achieved using the ER4043 filler wire.

In a study by Okokpujie et al. [9], MCDM techniques were applied for material selection in the development of horizontal wind turbine blades for sustainable energy applications. Using a quantitative approach based on AHP and TOPSIS, aluminium alloy was ranked as the most suitable material, followed by glass fibre, stainless steel, and mild steel. Thirumalai et al. [10] conducted an experimental study on the multi-response optimization of machining parameters for Inconel 718 using the TOPSIS approach. Their work focused on optimizing cutting conditions with carbide tools to simultaneously minimize cutting forces, maximize tool life, and enhance material removal rate, supported by NSGA-II optimization. Seo et al. [11] investigated the high-temperature behaviour of Ti-6Al-4V alloy by developing a constitutive model using the split Hopkinson pressure bar (SHPB) technique. The study emphasized the suitability of the alloy for high-strength applications such as aerospace components, including aircraft frames and turbine blades. Abbas et al. [12] applied a hybrid TOPSIS-fuzzy approach to optimize high-speed machining parameters of Ti-6Al-4V alloy. Their analysis considered multiple factors such as cutting speed, feed rate, and depth of cut, to demonstrate the material's excellent properties, including low density, high hardness, superior corrosion resistance, and fracture toughness, rendering it suitable for diverse industrial applications. Some of the relevant previous studies are summarized in Table 1.

Despite extensive research on material selection for cryogenic applications using various MCDM techniques such as AHP, TOPSIS, VIKOR, and COPRAS, several limitations persist. Most existing studies worked on subjective weighting approaches, which might introduce bias and affect the reliability of decisions. Furthermore, limited attention has been given to objective weighting methods that could account for intercriteria correlation and data variability. Prior works often considered a restricted set of evaluation criteria and lacked a comprehensive integration of mechanical, thermal, and sustainability aspects. The application of advanced distance-based ranking methods,

such as Combinative Distance-based Assessment (CODAS), in cryogenic material selection remains relatively unexplored. Moreover, there is a noticeable absence of a unified and systematic framework that integrates objective weighting with robust ranking techniques to enhance transparency, consistency, and computational efficiency.

Table 1. Summary of previous literature

Method / Approach	Weighting Type	Ranking Technique	Key Features	Limitations	Ref.
AHP	Subjective	Pairwise comparison	Incorporates experts' judgment, simple, and widely used	High subjectivity and issues related to inconsistency	[5, 9]
TOPSIS	Subjective / Hybrid	Distance from ideal solution	Implements with ease, efficient ranking	Sensitivity to weight assignment	[4, 10]
VIKOR	Subjective / Hybrid	Compromise ranking	Focuses on ranking and selection with conflicting criteria	Subjective inputs required	[1]
COPRAS	Subjective	Proportional assessment	Handles beneficial and non-beneficial criteria effectively	Dependent upon assigned weights	[1]
EVAMIX	Subjective	Mixed data evaluation	Considers to be suitable for both qualitative and quantitative data	Computational complexity	[1]
AHP–TOPSIS (Hybrid)	Subjective + Objective	Distance-based ranking	Combines structured weighting and ranking	Continuous influence from subjective judgments	[5, 9]
TOPSIS–Fuzzy	Hybrid	Fuzzy distance-based ranking	Handles uncertainty and vagueness	Increased computational complexity	[12]
WPM	Subjective	Weighted scoring	Possesses simple and fast computation	Limited handling of intercriteria relationships	[6]
CRITIC–CODAS (Proposed)	Objective	Distance from negative ideal (Euclidean + Taxicab)	Eliminates subjectivity in weighting; considers intercriteria correlation, robust ranking, and high transparency	Requires computational steps, a relatively new approach	Present Study

Note: VIKOR: VlseKriterijumska Optimizacija I Kompromisno Resenje; AHP: Analytic Hierarchy Process; TOPSIS: Technique for Order of Preference by Similarity to Ideal Solution; CRITIC: Criteria Importance through Intercriteria Correlation; CODAS: Combinative Distance-based Assessment.

To address these gaps, the present study proposed an integrated MCDM framework combining the Criteria Importance through Intercriteria Correlation (CRITIC) and CODAS methods for material selection. The primary objectives of this study are as follows:

- To develop an integrated CRITIC–CODAS framework for material selection;
- To evaluate suitable materials for cryogenic tank applications based on multiple technical criteria;
- To reduce subjectivity by determining objective criteria weights using the CRITIC method;
- To rank alternative materials using the CODAS method based on their distance from the negative ideal solution;

and

- To identify the most optimal material for liquid nitrogen cryogenic storage applications.

The novelty of this study lied in the integration of the CRITIC method for objective weighting with the CODAS method for robust ranking. This combined approach minimized subjective bias, improved ranking reliability, and enhanced the overall decision-making process. Therefore, this study provided a reliable, transparent, and computationally efficient decision-support framework for handling problems related to advanced engineering material selection.

2 Materials and Method

2.1 Materials

The selection of candidate materials and corresponding property data in this study was based on the work of Chatterjee et al. [1], which was widely recognized in the literature for cryogenic material evaluation. This dataset has been adopted due to its completeness, consistency, and relevance to cryogenic storage applications, particularly for liquid nitrogen. As shown in Table 2, the seven selected materials (Al2024-T6, Al5052-O, SS301-FH, SS301-AH, Ti-6Al-4V, Inconel, and 70Cu-30Zn) represent a diverse range of engineering alloys, including aluminum alloys, stainless steels, titanium alloys, superalloys, and brass, which are commonly used or considered for low-temperature applications. This diversity ensured comprehensive analysis across different material classes. Wherein, the selection of evaluation criteria was guided by their critical relevance to cryogenic tank performance. The seven criteria encompassing toughness index (TI), yield strength (YS), Young's modulus (YM), density (D), thermal expansion (TE), thermal conductivity (TC), and specific heat (SH) are widely acknowledged as key parameters influencing structural integrity, thermal stability, and operational efficiency at extremely low temperature. Mechanical properties such as toughness, yield strength, and modulus ensure structural reliability under cryogenic stresses, while thermal properties govern heat transfer and dimensional stability. Density is an important factor in reducing overall system weight, particularly in transport applications. The inclusion of these criteria provided a balanced and application-oriented evaluation framework.

Table 2. Properties of different selected materials (decision matrix) [1]

Material	Criteria						
	TI	YS	YM	D	TE	TC	SH
AL2024-T6	75.5	420	74.2	2.8	21.4	0.37	0.16
AL5052-O	95	91	70	2.68	22.1	0.33	0.16
SS301-FH	770	1365	189	7.9	16.9	0.04	0.08
SS301-AH	187	1120	210	7.9	14.4	0.03	0.08
TI-6AL-4V	179	875	112	4.43	9.4	0.016	0.09
INCONEL	239	1190	217	8.51	11.5	0.31	0.07
70CU-30ZN	273	200	112	8.53	19.9	0.29	0.06

Note: TI = toughness index, YS = yield strength, YM = Young's modulus, D = density, TE = thermal expansion, TC = thermal conductivity, SH = specific heat.

2.2 Methodology

An integrated framework based on CRITIC and CODAS, was utilized in the present study as it could overcome limitations of traditional MCDM approaches. The CRITIC method enabled objective determination of criteria weights by considering both data variability and intercriteria correlation, thereby reducing subjectivity. The CODAS method, on the other hand, provided a robust ranking mechanism based on the distance from the negative ideal solution using both Euclidean and Taxicab measures. A combination of these methods helped ensure improved accuracy, transparency, and reliability in decision making. The detailed information and steps of the process are presented below.

2.2.1 Criteria Importance through Intercriteria Correlation method

The CRITIC method is an objective MCDM technique used to determine the weight of criteria by considering both the contrast intensity (measured through standard deviation) and the level of conflict (measured through correlation) among the criteria. A higher standard deviation indicates greater discriminating power, while lower correlation between criteria suggests higher informational value. By combining these factors, CRITIC objectively assigns weights without relying on subjective judgments. It is especially useful when decision-makers are absent or when minimizing bias is essential. The method has been effectively applied in fields such as pharmaceuticals, water resource management, logistics, and organizational performance evaluation, hence demonstrating its wide applicability and reliability. This method uses standard deviation for measuring the contrast intensity of each criterion. It helps guarantee the criterion with a greater contrast intensity or standard deviation will be assigned with a higher weight.

The steps for the same is:

Step 1: Normalize the decision matrix

$$\overline{X}_{ij} = \frac{x_{ij} - x_j^{\text{worst}}}{x_j^{\text{best}} - x_j^{\text{worst}}} \quad (1)$$

Step 2: Calculate standard deviation σ_j for each criterion.

Step 3: Determine the symmetric matrix of $m \times n$ with element r_{jk} , which linear correlation coefficient between the vector x_j

Step 4: Calculate measure of the conflict created by criterion with respect to the decision situation defined by the rest of criteria.

$$\sum_{k=1}^m (1 - r_{jk}) \quad (2)$$

Step5: Determine the quantity of the information in relation to each criterion.

$$C_j = \sigma_j * \sum_{k=1}^m (1 - r_{jk}), k = 1, 2, 3 \dots m \quad (3)$$

where, σ_j is the standard deviation of the j^{th} criterion, m is, the number of criteria, r_{jk} is the Pearson correlation coefficient between criteria i and k .

Step 6: Determine the objective weight

$$W_j = \frac{C_j}{\sum_{k=1}^m C_k} \quad (4)$$

where, W_j is the weight of k^{th} criteria.

The Pearson correlation coefficient, limited to linear relationships, may not accurately reflect independence between criteria, especially with positive correlations. In the CRITIC method, strong negative correlations could lead to redundant weighting as negatively correlated criteria may receive high weights, thus skewing the results. Using the absolute value of correlations could reduce this redundancy; due to Pearson's linear limitation, a measure capturing both linear and nonlinear dependencies is recommended for more accurate weight assignment [13, 14].

2.2.2 Combinative Distance-Based Assessment (CODAS) method

This section introduces the CODAS method, which evaluates alternatives based on their distance from the negative ideal solution using two metrics: The primary measure is the Euclidean distance (l_2 -norm), and the secondary is (l_1 -norm). Alternatives farther from the negative ideal are considered more desirable. When alternatives are indistinguishable using the Euclidean distance, the Taxicab distance serves as a tiebreaker. Although the method favours the l_2 -norm indifference space, both l_1 and l_2 norms are incorporated. The procedure for applying CODAS with n alternatives and m criteria aims to perform multi-criteria selection process, in accordance with the largest Euclidean and Taxicab distance with respect to negative ideal solutions [15–17]. The steps for the CODAS are outlined below:

Step 1: Develop the initial decision-making matrix (X)

$$X = [x_{ij}]_{n \times m} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \quad (5)$$

where, n is the number of alternatives and m is the number of criteria. x_{ij} ($x_{ij} \geq 0$) represents the performance score of the i^{th} alternative with respect to the j^{th} criterion, where $i \in \{1, 2, \dots, n\}$ and $j \in \{1, 2, \dots, m\}$.

Step 2: The normalized decision matrix is calculated using linear normalization of the performance values, as follows:

$$n_{ij} = \begin{cases} \frac{x_{ij}}{\max x_{ij}} & \text{if } j \in N_b \\ i & \\ \min x_{ij} & \\ \frac{i}{x_{ij}} & \text{if } j \in N_c \end{cases} \quad (6)$$

where, N_b represents the benefit it, and N_c represents the cost criteria.

Step 3: Compute the weighted normalized decision matrix using the following formula for weighted performance values:

$$r_{ij} = w_j \times n_{ij} \quad (7)$$

where, w_j ($0 < w_j < 1$) denotes the weight of j^{th} criterion.

Step 4: Determine the negative ideal solution (point) as follows:

$$\begin{aligned} ns &= [n_{sj}]_{1 \times m} \\ n_{sj} &= \min_i r_{ij} \end{aligned} \quad (8)$$

Step 5: Calculate the Euclidean and Taxicab distance of alternatives from the negative ideal solution

$$E_i = \sqrt{\sum_{j=1}^m (r_{ij} - n_{sj})^2} \quad (9)$$

$$T_i = \sum_{j=1}^m |r_{ij} - n_{sj}| \quad (10)$$

Step 6: Construct the relative assessment matrix

$$R_a = [h_{ik}]_{n \times n} \quad (11)$$

$$h_{ik} = (E_i - E_k) + \{\phi(E_i - E_k) \times (T_i - T_k)\} \quad (12)$$

where, $k \in \{1, 2, \dots, n\}$ and ϕ denote a threshold function to recognize the equality of the Euclidean distances of two alternatives.

Step 7: Calculate the assessment score & rank the alternatives

$$H_i = \sum_{k=1}^n h_{ik} \quad (13)$$

Step 8: Rank the alternatives in descending order based on their assessment values.

The alternative with the highest H_i is the best choice among all the alternatives.

3 Results and Discussion

To demonstrate applicability of the two proposed frameworks, a case of benchmarked data from literature had been considered for the selection of material for cryogenic storage tanks used in the transportation of liquid nitrogen [1]. This study evaluated seven candidate materials against seven selection criteria: Toughness Index (TI), Yield Strength (YS), Young's Modulus (YM), Density (D), Thermal Expansion Coefficient (TE), Thermal Conductivity (TC), and Specific Heat (SH). Among these, TI, YS, and YM were considered beneficial criteria, in which higher values indicated better performance. In contrast, D, TE, TC, and SH were non-beneficial since lower values were preferred.

3.1 Criteria Importance through Intercriteria Correlation Method

The problem of material selection for the design of cryogenic storage tanks was first solved by the CRITIC method. The very first step involved identifying the best and worst values for each criterion based on whether the attribute was beneficial or not. As regards beneficial criteria, the highest values were considered best, while for non-beneficial criteria, the lowest values were preferred. These selected values are summarized in Table 3, which lists the best and worst performance values for each criterion. These values were used to normalize the decision matrix using Eq. (1). This normalization process ensured all criteria were brought to a comparable scale. Based on these normalized values, the normalized decision matrix was then computed, as shown in Table 4.

Table 4 highlights the relative performance of each material across multiple criteria on a standardized scale. Among the materials, SS301-FH and SS301-AH showed consistently high weighted scores across most criteria, thus indicating strong overall performance, while AL2024-T6 and AL5052-O exhibited lower values, suggesting comparatively weaker performance in this multi-criteria evaluation. The next step was to discover the standard deviation values, as shown in Table 5.

Once the standard deviation (σ_k) for each criterion was calculated, the next step was to evaluate the degree of interdependence between criteria using linear correlation. Table 6 presents the symmetric matrix of Pearson correlation coefficient, which quantifies the linear relationship between each pair of criteria. Diagonal elements are 1, representing perfect correlation of each criterion with itself. Off-diagonal values indicate the strength and direction of correlation (positive) values to suggest a direct relationship, while negative values indicate an inverse

relationship. For example, a strongly negative correlation was observed between ductility (D) and shear strength (SH) at -0.924, implying these two criteria behaved oppositely. Thereafter, in Table 7, measure of the conflict created by the criteria needed to be calculated with respect to the decision defined by the rest of the criteria using Eq. (2).

Finally, the amount of information for each criterion (C_j) was calculated using Eq. (3). Based on these values, the objective weights (W_j) were determined using Eq. (4), as listed in Table 8.

Table 3. Identification of best and worst values for each criterion in the Criteria Importance through InterCriteria Correlation (CRITIC) method

Best	770	1365	217	2.68	9.4	0.016	0.06
Worst	75.5	91	70	8.53	22.1	0.37	0.16

Table 4. Normalized decision matrix using the Criteria Importance through InterCriteria Correlation (CRITIC) method

Material	Criteria						
	TI	YS	YM	D	TE	TC	SH
AL2024-T6	0	0.258242	0.028571	0.979487	0.055118	0	0
AL5052-O	0.028078	0	0	1	0	0.112994	0
SS301-FH	1	1	0.809524	0.107692	0.409449	0.932203	0.8
SS301-AH	0.160547	0.807692	0.952381	0.107692	0.606299	0.960452	0.8
TI-6AL-4V	0.149028	0.615385	0.285714	0.700855	1	1	0.7
INCONEL	0.235421	0.862637	1	0.003419	0.834646	0.169492	0.9
70CU-30ZN	0.284377	0.085557	0.285714	0	0.173228	0.225989	1

Note: TI = toughness index, YS = yield strength, YM = Young's modulus, D = density, TE = thermal expansion, TC = thermal conductivity, SH = specific heat.

Table 5. Standard deviation (σ) of each criterion from the weighted normalized matrix

Standard Deviation	0.339655	0.40152	0.430467	0.46063	0.389623	0.453047	0.420317
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Table 6. Symmetric matrix of linear correlation coefficients between criteria in the Criteria Importance through InterCriteria Correlation (CRITIC) method

Property	TI	YS	YM	D	TE	TC	SH
TI	1	0.593576	0.489701	-0.54433	0.106581	0.476866	0.483953
YS	0.593576	1	0.873525	-0.55992	0.71087	0.664615	0.554167
YM	0.489701	0.873525	1	-0.81615	0.597904	0.474873	0.716745
D	-0.54433	-0.55992	-0.81615	1	-0.37891	-0.31177	-0.92411
TE	0.106581	0.71087	0.597904	-0.37891	1	0.612535	0.600213
TC	0.476866	0.664615	0.474873	-0.31177	0.612535	1	0.483114
SH	0.483953	0.554167	0.716745	-0.92411	0.600213	0.483114	1

Note: TI = toughness index, YS = yield strength, YM = Young's modulus, D = density, TE = thermal expansion, TC = thermal conductivity, SH = specific heat.

3.2 Combinative Distance-based Assessment Method

The same material selection problem for cryogenic storage tanks was solved using CODAS method. In this method, the same normalized decision matrix and weights of criteria were considered (refer to Table 4 and Table 8, respectively). Thereafter, weighted normalized decision matrix, represented in Table 9, was calculated using Eq. (7).

Negative ideal solution was obtained by applying Eq. (8) to the values highlighted in Table 9 and was determined as shown in Table 10.

As regards the CODAS method, there was a need to calculate the Euclidean and Taxicab distances of alternatives from the negative ideal solution using Eqs. (9) and (10), respectively. The calculation results are represented in Table 11.

Table 7. Intercriteria conflict matrix for information content evaluation in the Criteria Importance through Intercriteria Correlation (CRITIC) method

Property	TI	YS	YM	D	TE	TC	SH
TI	0	0.406424	0.510299	1.544328	0.893419	0.523134	0.516047
YS	0.406424	0	0.126475	1.55992	0.28913	0.335385	0.445833
YM	0.510299	0.126475	0	1.81615	0.402096	0.525127	0.283255
D	1.544328	1.55992	1.81615	0	1.378907	1.311768	1.924109
TE	0.893419	0.28913	0.402096	1.378907	0	0.387465	0.399787
TC	0.523134	0.335385	0.525127	1.311768	0.387465	0	0.516886
SH	0.516047	0.445833	0.283255	1.924109	0.399787	0.516886	0

Note: TI = toughness index, YS = yield strength, YM = Young's modulus, D = density, TE = thermal expansion, TC = thermal conductivity, SH = specific heat.

Table 8. Information content and objective weights for each criterion using the Criteria Importance through Intercriteria Correlation (CRITIC) method

Sum (1-RJK)	Sigma (Std. Dev.)	C_j	W_j	Property
4.393652	0.339655	1.492326	0.110206	TI
3.163167	0.40152	1.270075	0.093793	YS
3.663403	0.430467	1.576974	0.116457	YM
9.535181	0.46063	4.39219	0.324357	D
3.750804	0.389623	1.461399	0.107922	TE
3.599766	0.453047	1.630863	0.120437	TC
4.085918	0.420317	1.717381	0.126826	SH
Sum of C_j		13.54121		

Table 9. Weighted normalized decision matrix for the Combinative Distance-based Assessment (CODAS) method

Material	TI	YS	YM	D	TE	TC	SH
AL2024-T6	0	0.024221	0.003327	0.317704	0.005948	0	0
AL5052-O	0.003094	0	0	0.324357	0	0.013609	0
SS301-FH	0.110206	0.093793	0.094275	0.034931	0.044189	0.112272	0.101461
SS301-AH	0.017693	0.075756	0.110912	0.034931	0.065433	0.115674	0.101461
TI-6AL-4V	0.016424	0.057719	0.033274	0.227327	0.107922	0.120437	0.088778
INCONEL	0.025945	0.08091	0.116457	0.001109	0.090077	0.020413	0.114144
70CU-30ZN	0.03134	0.008025	0.033274	0	0.018695	0.027217	0.126826

Note: TI = toughness index, YS = yield strength, YM = Young's modulus, D = density, TE = thermal expansion, TC = thermal conductivity, SH = specific heat.

Table 10. Determination of the negative ideal solution

Negative Ideal	0	0	0	0	0	0	0
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Table 11. Euclidean (EI) and Taxicab (TI) distances of alternatives from the negative ideal solution in the Combinative Distance-based Assessment (CODAS) method

Material	EI	TI
AL2024-T6	0.318698713	0.35120104
AL5052-O	0.324657444	0.34106039
SS301-FH	0.236437059	0.59112691
SS301-AH	0.218013657	0.52186029
TI-6AL-4V	0.300699187	0.65188149
INCONEL	0.205772634	0.44905445
70CU-30ZN	0.139028508	0.24537727

Subsequently, it was required to construct the relative assessment matrix using Eq. (11) and the results are presented in Table 12.

Table 12. Relative assessment matrix of alternatives in the Combinative Distance-based Assessment (CODAS) method

Material	1	2	3	4	5	6	7
AL2024-T6	0	-0.00596	0.0818669	0.1003414	0.0178913	0.1127051	0.1800505
AL5052-O	0.0059575	0	0.0877792	0.1062582	0.0238093	0.118628	0.1859842
SS301-FH	-0.082656	-0.088662	0	0.0184489	-0.064184	0.0307516	0.0980821
SS301-AH	-0.101029	-0.107029	-0.018398	0	-0.082471	0.0122589	0.0794219
TI-6AL-4V	-0.018108	-0.024107	0.0643402	0.0829006	0	0.0953116	0.1629851
INCONEL	-0.113147	-0.119141	-0.030577	-0.012223	-0.094541	0	0.067016
70CU-30ZN	-0.17929	-0.185273	-0.096735	-0.078548	-0.160356	-0.066472	0

Finally, assessment scores of alternatives were calculated using Eq. (12) and the alternatives were ranked according to decreasing values of assessment as shown in Table 13.

Table 13. Assessment scores (HI) and final ranking of alternatives in the Combinative Distance-based Assessment (CODAS) method

Material	HI	Rank
AL2024-T6	0.486895	2
AL5052-O	0.528416	1
SS301-FH	-0.08822	4
SS301-AH	-0.21725	5
TI-6AL-4V	0.363323	3
INCONEL	-0.30261	6
70CU-30ZN	-0.76668	7

As observed from Table 13, Al5052-O achieved the highest assessment score among all alternatives, followed by Al2024-T6, Ti-6Al-4V, SS301-FH, SS301-AH, Inconel 718, and 70Cu-30Zn. Accordingly, Al5052-O was identified as the most suitable material for cryogenic tank applications involving storage of liquid nitrogen. While the present investigation utilized benchmark data from the literature to establish methodological validity, the proposed framework demonstrated broad generalizability and was readily extendable to industrial cryogenic applications. The selected case study was designed to emulate realistic engineering conditions and incorporate materials commonly employed in practice, thereby ensuring methodological rigor and underscoring the industrial relevance of the findings.

4 Sensitivity Analysis

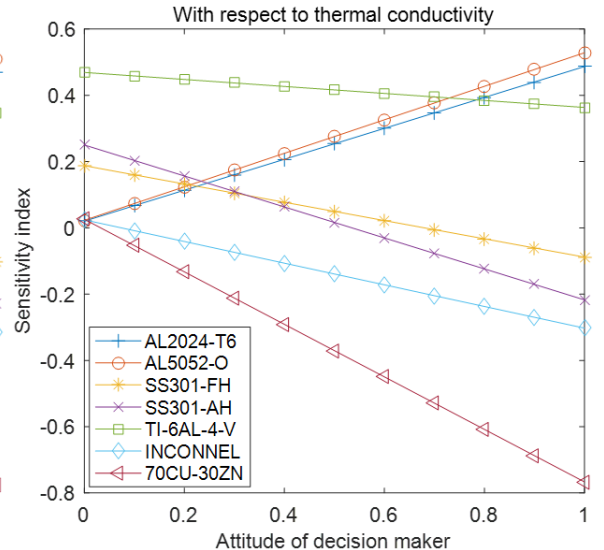
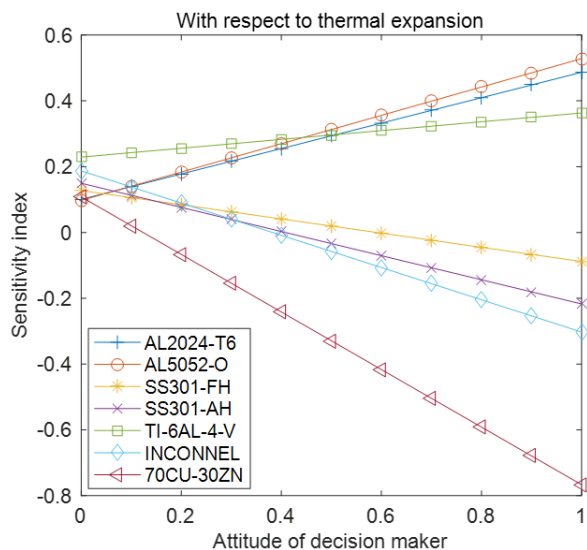
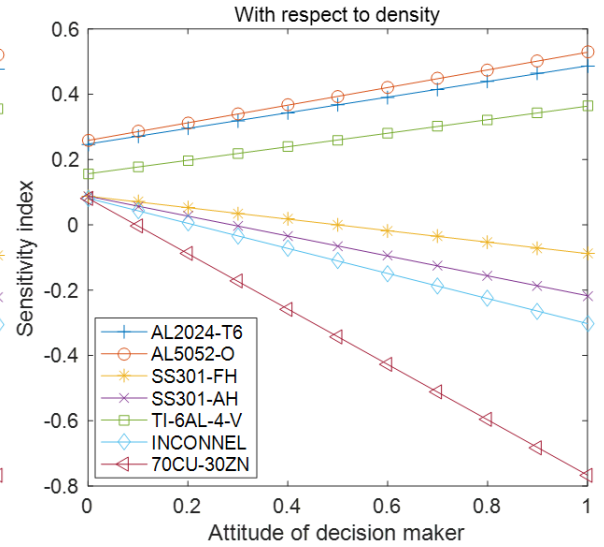
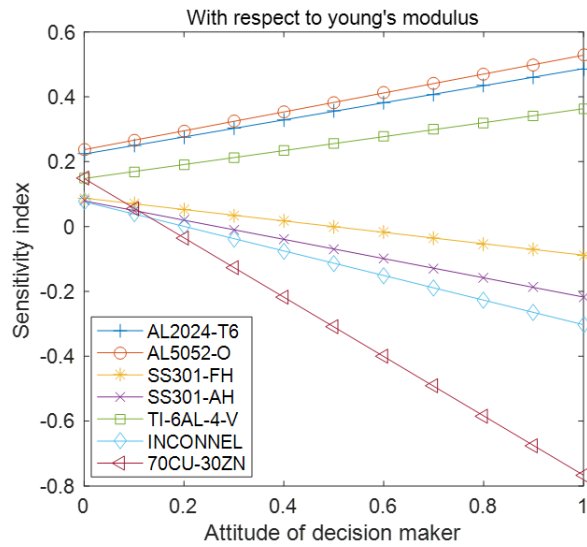
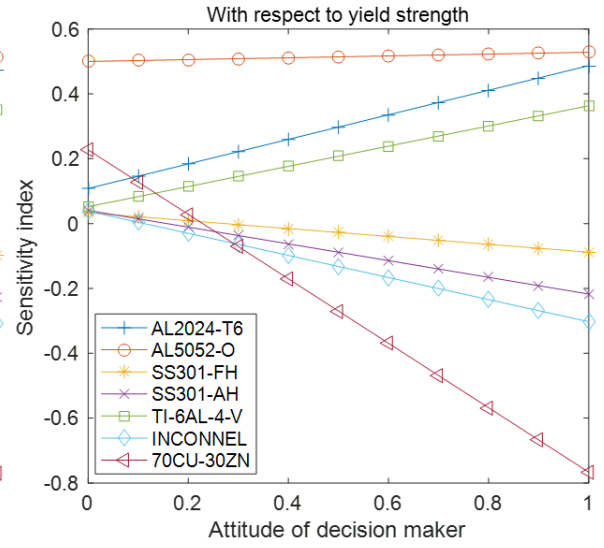
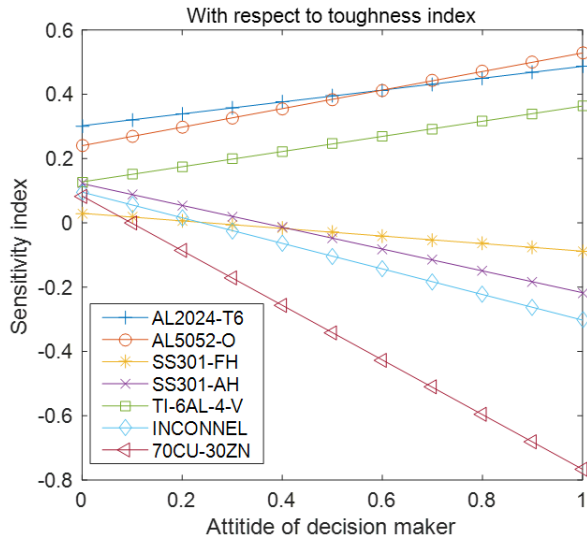
Sensitivity analysis is a systematic approach used to evaluate how uncertainty in the output of a model could be attributed to different sources of uncertainty in its input variables. It plays a crucial role in validating decision-making models by examining the robustness and reliability of the obtained results under varying conditions. In the present study, a mathematical framework was employed to investigate the uncertainty associated with selecting the most suitable chitosan-based biodegradable biopolymer composite for energy storage device application. This evaluation was conducted across three distinct segments, having considered multiple criteria and alternative rankings, as supported by earlier studies [18–20].

$$SI_i = [(\alpha SFM_i) + (1 - \alpha) OFM_i] \quad (14)$$

where,

$$OFM_i = \frac{1}{[OFD_i \sum_{i=1}^n OFD_i^{-1}]} \quad (15)$$

In this framework, the sensitivity index was denoted by SI, while α (alpha) represents the weight assigned to the objective factor. The global priority values of each alternative were derived by using the Simple Multi-Attribute Rating Technique (SMART-based Structure-from-Motion (SFM) positions), and the Objective Factor Measure (OFM) reflected obtained quantitative data. The total number of alternatives considered in this analysis was $n = 7$, and the dimensionality of the objective factors was represented by OFD. The normalized SFM values obtained from Table 13, were utilized as input in Eq. (14), thus forming the basis for sensitivity computations.



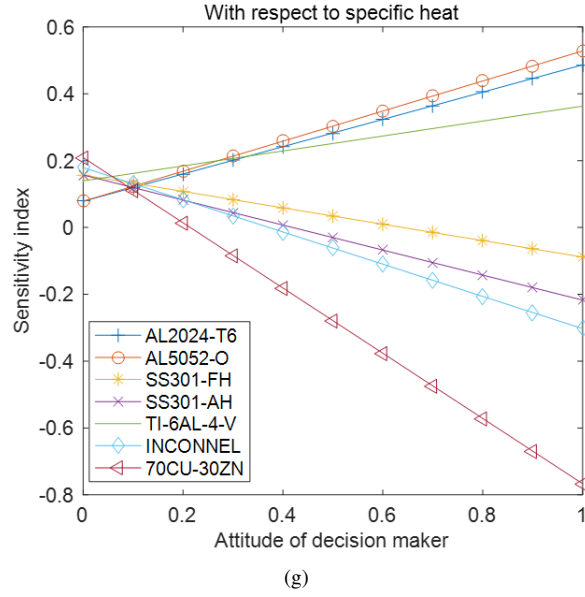


Figure 1. Sensitivity analysis of alternatives under various criteria

A critical aspect of this analysis was the selection of the parameter α , which signified the relative importance assigned to objective factors compared with subjective considerations. The choice of α was inherently dependent on the judgment, preference, and perception of the decision maker regarding the significance of different evaluation criteria. As a result, varying α values could lead to different ranking outcomes for the same set of alternatives, thus highlighting the need for a comprehensive sensitivity assessment.

To address this, the study evaluated multiple scenarios by systematically varying α , thereby generating a range of alternative rankings. The results are visually represented via sensitivity plots in Figure 1(a–g), illustrating how changes in α influenced the ranking stability of the considered cryogenic tank material. These graphical representations provide valuable insights into the consistency and robustness of the decision making process.

The outcomes, summarized in Table 14, emphasized the importance of carefully selecting an appropriate value of α , as it significantly affected the final decision. The analysis demonstrated that, despite variations in α , the ranking patterns remained largely consistent across different scenarios. This consistency confirmed the reliability and stability of the integrated decision-making approach employed in the study.

To sum up, sensitivity analysis not only validated the robustness of the selected optimal material but also reinforced confidence in the integrated framework used. It ensured that the final decision was not unduly influenced by minor variations in input parameters, thereby supporting a more informed and dependable material selection process for cryogenic tanks.

Table 14. Analysis of Figure 1(a)–(g)

Ref. of Figure	Ranking	Variation of Attitude
1(a)	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.589$
	SS301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.412$
	SS301-FH > INCONEL > 70CU-30ZN	$\alpha = 0.226$
1(b)	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.097$
	S70CU-30ZN > 301-FH > SS301-AH > INCONEL	$\alpha = 0.154$
	S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.215$
	SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.243$
	INCONEL > 70CU-30ZN	$\alpha = 0.293$
1(c)	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.097$
	SS301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.114$
	INCONEL > 70CU-30ZN	$\alpha = 0.195$

Ref. of Figure	Ranking	Variation of Attitude
1(d)	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	At any α value, the ranking is same as integrated framework
	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.426$
	AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.441$
	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.115$
1(e)	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.087$
	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.079$
	S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.227$
	S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.178$
	SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.245$
	AL5052-O > TI-6AL-4V > AL2024-T6 > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.676$
	AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.791$
1(f)	AL5052-O > AL2024-T6 > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.211$
	AL5052-O > AL2024-T6 > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.232$
	AL5052-O > AL2024-T6 > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.248$
	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.234$
	AL5052-O > AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.311$
1(g)	AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.187$
	AL2024-T6 > TI-6AL-4V > S301-FH > SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.198$
	SS301-AH > INCONEL > 70CU-30ZN	$\alpha = 0.208$

5 Conclusions

The present study demonstrated the applicability, capability, and accuracy of both the CRITIC and CODAS methods in addressing complex material selection problems of cryogenic tanks. Using the CRITIC method, a decision maker could evaluate alternatives and determine the weighted criteria w_j for each factor without requiring a deep understanding of the underlying decision-making process. In contrast, the CODAS method evaluated the performance of alternatives in a multi-criteria selection process by calculating the largest Euclidean and Taxicab distances from the negative ideal solution, thereby identifying the material with the highest assessment score. Both methods enabled a more effective evaluation of material alternatives. In this case, density was found to have the highest weight, followed by specific heat, thermal conductivity, Young's modulus, toughness index, thermal expansion, and yield strength. The optimal material for the cryogenic tank application was determined to be AL5052-O. Although the present study utilized benchmark data from the literature for methodological validation, the proposed framework was generic and could be readily applied to real industrial scenarios involving cryogenic storage systems. The selected case reflected practical engineering conditions and commonly used materials, thereby ensuring the applicability of the results. Future work may involve validation using real-time industrial data to further enhance the robustness and practical relevance of the proposed approach.

Author Contributions

Conceptualization, C.B. and K.C.; methodology, C.B., P.K., S.D.L., P.D., and C.M.; software, C.B., C.M., and A.K.; validation, C.B., C.M., and K.C.; formal analysis, C.B. and C.M.; investigation, C.M. and C.B.; resources,

C.B.; data curation, S.D.L. and P.D.; writing—original draft preparation, C.B. and P.K.; writing—review and editing, C.B. and K.C.; visualization, C.B. and C.M.; supervision, K.C. and A.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work the authors used GPT-4 & Google Gemini 3.0 to improve readability, rectify grammatical errors, and develop schematics. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

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